

Home Assignment 1

Question 1

A researcher wants to check if the type of clothing store (a total of 4 types) affects the expected price of the clothes. Eight clothing items were randomly sampled from each store to test the hypothesis. The results can be found in the clothes spreadsheet in the file class.exe.01

- A. Present two theoretical models – ANOVA and regression- and detail their components.
- B. Write a suitable hypothesis for each model and conclude by citing the test statistic and p-value. Comment on the similarities and differences between the hypothesis and the tests.
- C. Calculate the average prices in each store using the linear regression coefficients, with store number 4 as the control group.
- D. Calculate the effects of the different groups using the grand average obtained in section A and the sample averages calculated in section C.

Question 2

Let the model be

$$Y_i = \beta I\{X_i > c\} + \epsilon_i, \quad i = 1, \dots, n$$

where $\epsilon_i \sim^{iid} N(0, \sigma^2)$, and $c > 0$.

- A. Derive the OLS estimator of β both in matrix and direct forms and compare it to the MLE estimator.
- B. Compute the expected value and the variances both in matrix and in the direct form of the estimator.
- C. Let

$$c = 2$$

$$x = (1, 0, -1, 3, 4, 5), \quad y = (10, 11, 12, 9, 10, 9)$$

Compute the point estimator of β and its estimated variance.

- D. Check the model significance. Present the test statistic and the p-value.
- E. Compute the 95% confidence interval.
- F. Repeat the calculations in Python. Comment on the differences and the similarities between your manual computations and the “automatic” ones.

Question 3

An investment portfolio manager wants to check whether there are differences in the expected annual return in investment portfolios in different investment strategies (A, B, C, D). The data is in class.exe.01, sheet named invest.

- A. Check visually whether the returns are normally distributed: Draw histograms of the distribution of returns for each strategy.
- B. Estimate the appropriate ANOVA test.

C. Perform post hoc tests with Fisher's LSD, Bonferroni, and FDR corrections for p-values. Summarize your findings and emphasize the differences between the various methods.

D. Do you think the parametric normal model was suitable for the given data? If not, can you suggest an alternative approach?

Question 4

Use the rats.csv file. This file contains data from a toxicological experiment on mice. The 48 mice were randomly allocated to receive three types of poisons (I, II and III) and four treatment groups (A, B, C, D). The dependent variable is the response time to a given stimulus.

A. Is the experiment balanced? Plot Box-Plots. Discuss the results.

B. Write down the theoretical ANOVA model and list its constraints and assumptions.

c. Estimate the two-way ANOVA model (without interaction) and summarise the findings. Write down the hypothesis being tested and the test statistic.

d. Write a suitable regression model. Write down the assumptions of the model.

e. Evaluate the model and compare the findings to section c.

f. Check the model assumptions graphically.

g. Perform the Fisher, Bonferroni, and FDR post hoc tests. Summarise the findings.

שאלה 5

להלן נתונים שנאספו ממחקר פורץ דרך על הקשר בין x ל- y

$$\bar{x} = 5, \quad \bar{y} = 3, \quad n = 20, \quad \sum (x_i - \bar{x})^2 = 160$$

$$\sum (y_i - \bar{y})^2 = 83.2, \quad \sum (x_i - \bar{x})(y_i - \bar{y}) = 80$$

החוקרים הניחו כי המודל $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ מתאר באופן מדויק את הקשר בין x ל- y .

א. אמדו את משוואת הרגרסיה.

ב. בנו את טבלת ניתוח השונות ובחנו את מובהקות המודל.